

Math 196 Problem Set 4

Due April 23, at 10:00 PM

For book problems, see the pdf on canvas or in my email.

1 Homework Problems

Problem 1. Consider the following linear subspaces of $\mathbb{R}^3$:

$$V = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}, \quad W = \text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

These are both planes through the origin, intersecting in a line. We would like to find a vector spanning this line.

- (a) Find equations defining V and W .
- (b) Put these together to get a system of equations defining $V \cap W$, and solve this system to find the required vector.

Problem 2. Consider a general 2×2 matrix

$$\begin{bmatrix} w & x \\ y & z \end{bmatrix} = A$$

- (a) Describe the matrix products

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix}$$

in terms of their effect on the rows of A .

- (b) Find matrices T , P_k , Q_k for $k \in \mathbb{R}$, $k \neq 0$ such that TA is obtained by swapping the rows of A , $P_k A$ is obtained by scaling the first row of A by a factor of k , and $Q_k A$ is obtained by scaling the second row of A by a factor of k . Observe that all the matrices that appear on the left in these products are invertible.
- (c) How can the reduced row echelon form of A be expressed in terms of A and these matrices?
- (d) Using part (c), explain why the row echelon form of A has the same kernel as A . In other words, if R is reduced row echelon form, show that $Rv = 0$ implies $Av = 0$, and that $Av = 0$ implies $Rv = 0$.
- (e) Think about an analogous description of row operations for matrices larger than 2×2 and convince yourself that this also tells you that the kernel of a matrix is equal to the kernel of its reduced row echelon form. No need to write anything down for this part.

Problem 3. Determine the set of all $t \in \mathbb{R}$ such that the vectors

$$\begin{bmatrix} 2 \\ -1 \\ t \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} -t \\ t \\ 1 \end{bmatrix}$$

are linearly independent.

Problem 4. For each matrix, find a basis of its image and kernel, and determine its rank.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 1 \\ 2 & 0 & 4 \end{bmatrix}, \begin{bmatrix} -1 & 2 & 0 \\ 4 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

Problem 5. In each part, either find an example of a linear transformation with the given properties, or argue that one cannot exist.

- (a) A linear transformation $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ whose kernel has dimension 1.
- (b) A linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ which is injective but not surjective.
- (c) A linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ which is surjective but not injective.
- (d) A surjective linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^2$.
- (e) An injective linear transformation $\mathbb{R}^4 \rightarrow \mathbb{R}^3$.
- (f) A linear transformation $\mathbb{R}^4 \rightarrow \mathbb{R}^3$ whose image and kernel both have dimension 2.
- (g) An invertible linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ which is not the identity.
- (h) An invertible linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^4$.

Problem 6. Suppose that $v_1, \dots, v_r$ span $\mathbb{R}^n$ and that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear. Prove that $T(v_1), \dots, T(v_r)$ span $\text{im}(T)$.

Problem 7. In each part, find the coordinates of the vector w in the basis v_1, v_2 for $\mathbb{R}^2$.

(a)

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, w = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

(b)

$$v_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, w = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

2 Review Problems

Problem 8. Exercises 7, 9, 10, 11, 13 from Section 2.3 of book, on page 85.

Problem 9. Exercises 16 to 20 from Section 2.4 of the book, on page 97.

Problem 10. Write the solution set of

$$\begin{cases} 2x_2 - x_3 + 3x_5 = 0 \\ -x_1 + 2x_2 + 4x_3 - x_4 = 0 \\ x_4 + x_5 = 0 \end{cases}$$

as the image of a matrix.

Problem 11. Find a system of equations which defines the linear subspace

$$V = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Use these equations to determine if the following vectors are contained in V :

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 8 \\ 1 \\ 8 \\ -5 \end{bmatrix}, \begin{bmatrix} -6 \\ 0 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ -3 \end{bmatrix}$$

Problem 12. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation. Prove that the *fixed point set* of T ,

$$\{v \in \mathbb{R}^n \mid T(v) = v\}$$

is a linear subspace of $\mathbb{R}^n$.

Problem 13. Prove that if v_1, v_2, v_3, v_4 is a linearly independent set of vectors, then so is v_1, v_3 .

Problem 14. Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation. Prove that $S(v) = \frac{1}{2}(v + T(v))$ is a linear transformation. If T corresponds to a matrix A , what is the matrix corresponding to S ?